Deep Learning Framework for Automated Plant Disease Detection Using Leaf Images

ABSTRACT

Agriculture plays a vital role in ensuring food security and supporting the global economy. Plant diseases significantly reduce crop yield and quality, leading to economic losses for farmers and affecting agricultural productivity. Traditional disease identification methods rely on manual inspection by agricultural experts, which is time-consuming, labor-intensive, and may not always be available in remote farming areas. Early and accurate detection of plant diseases is therefore essential to minimize crop damage and improve farming efficiency.

This project proposes a Deep Learning Framework for Automated Plant Disease Detection Using Leaf Images that utilizes Convolutional Neural Networks (CNNs) and transfer learning to automatically identify plant diseases from leaf images. The proposed system employs a pre-trained EfficientNet model, which is fine-tuned using labeled plant leaf datasets. The framework performs image preprocessing, including resizing, normalization, and data augmentation, before extracting important visual features for disease classification. The trained model predicts the corresponding disease category and provides a confidence score for each prediction.

The proposed framework will be evaluated for its ability to classify multiple plant diseases as well as healthy leaves across different crop species. The system will be evaluated using performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix to measure classification effectiveness. The resulting framework aims to provide farmers, agricultural researchers, and extension workers with a fast, reliable, and cost-effective tool for early plant disease detection, thereby supporting timely disease management, reducing crop losses, and promoting sustainable agricultural practices.

DATASETS

Dataset	URL Link / Purpose
PlantVillage Dataset	https://www.kaggle.com/datasets/emmarex/plantdisease Primary dataset for training and evaluation.
PlantVillage (Mendeley)	https://data.mendeley.com/datasets/tywbtsjrjv Alternative source for the PlantVillage dataset.
PlantVillage (GitHub)	https://github.com/spMohanty/PlantVillage-Dataset Official dataset repository / source.
PlantVillage (Hugging Face)	https://huggingface.co/datasets/mohanty/PlantVillage Alternative dataset access.
PlantDoc Dataset	https://github.com/pratikkayal/PlantDoc-Dataset Optional external dataset for testing real-world generalization.

APPROACH

1. Plant Leaf Images

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2. PlantVillage Dataset

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3. Image Preprocessing (Resize, Normalize, Augmentation)

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4. Dataset Splitting (Training / Validation / Testing)

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5. Transfer Learning Model (EfficientNet)

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6. Feature Extraction

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7. Disease Classification

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8. Healthy / Disease Prediction

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9. Confidence Score Generation

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10. Performance Evaluation (Accuracy, Precision, Recall, F1-Score, Confusion Matrix)

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11. Web-Based User Interface (Image Upload)

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12. Final Disease Prediction with Plant & Disease Information

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